

Project Management

4. Integration Management & Project planning

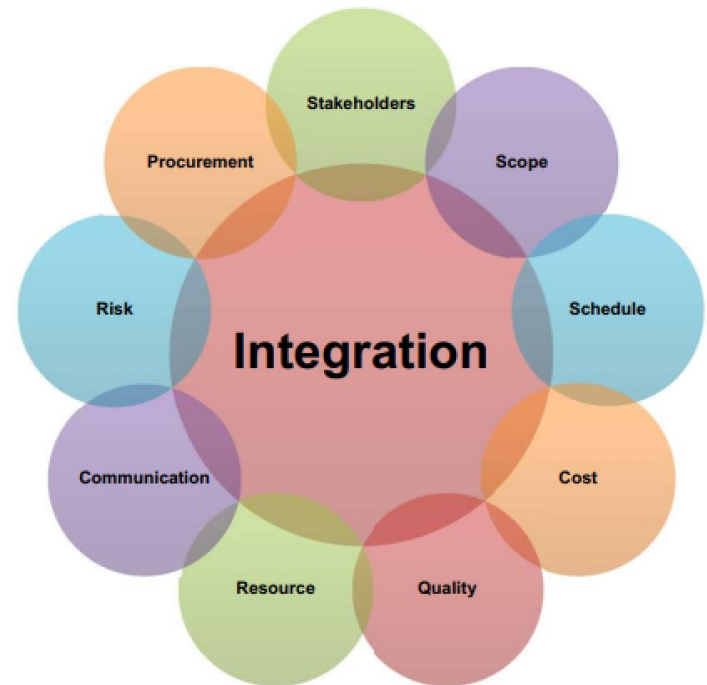
Contents

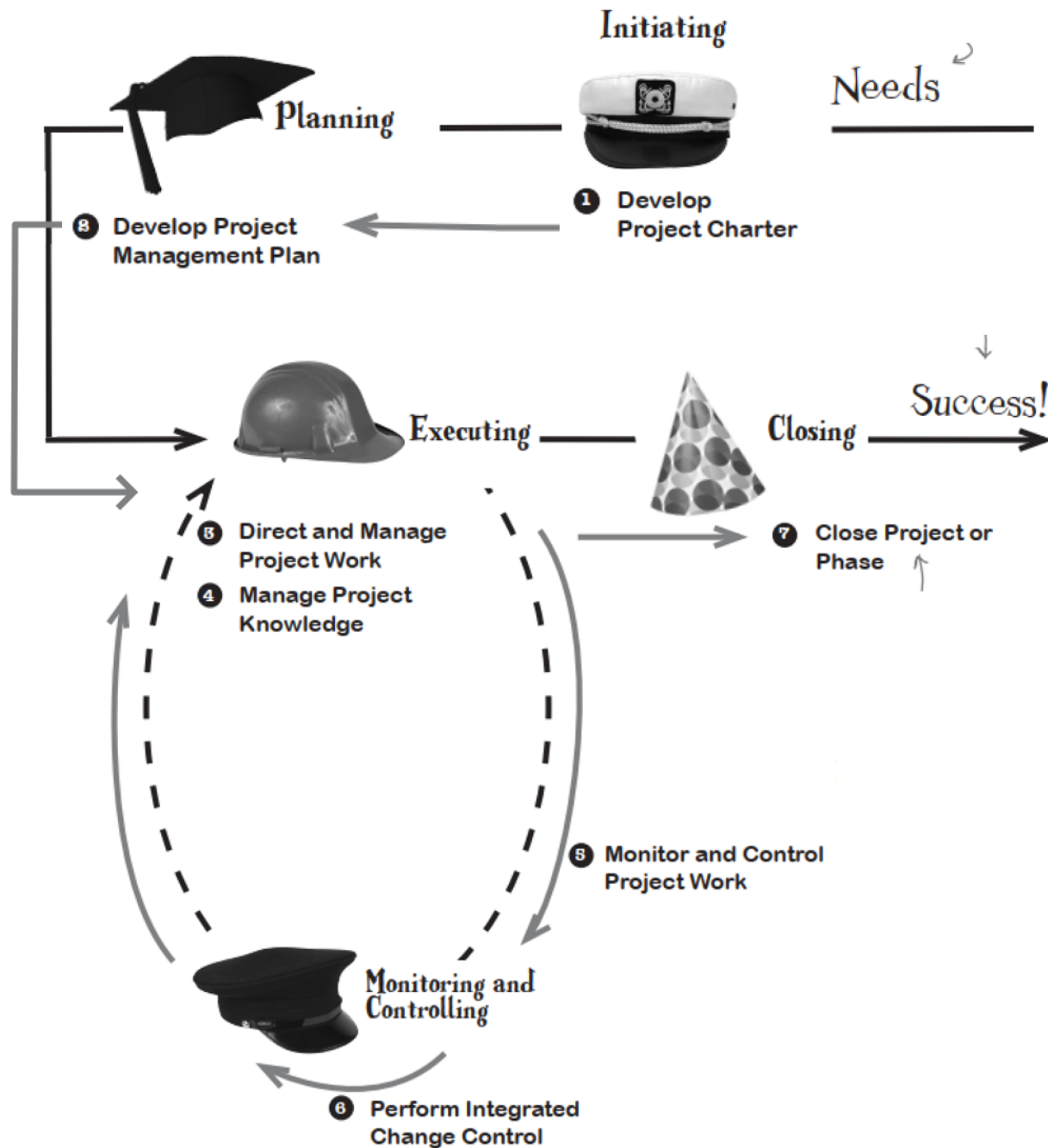
- Project Integration Management
- Develop Project Management Plan Process
 - Subsidiary plans
 - Baseline
- Stakeholder Management
- Plan Stakeholder Engagement Process
- Stakeholder engagement assessment matrix
- Communication Management
- Plan Communication Management
 - Communication model
 - Communication methods
 - Communication levels

Project Integration Management

Project Integration Management

- Day-to-day work of a **project manager**: make projects run well
 - Pull all the pieces of a project together into a **cohesive whole**.
- Includes characteristics of **unification, consolidation, communication, and interrelationship** of various processes and project management activities





Develop Project Management Plan

Develop Project Management Plan

- Defining, preparing, and coordinating all plan components and consolidating them into an integrated project management plan
 - Basis of all project work
 - How the work will be performed

Inputs

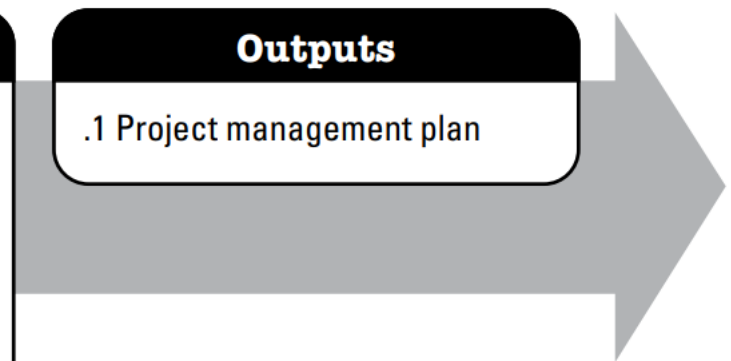
- .1 Project charter
- .2 Outputs from other processes
- .3 Enterprise environmental factors
- .4 Organizational process assets

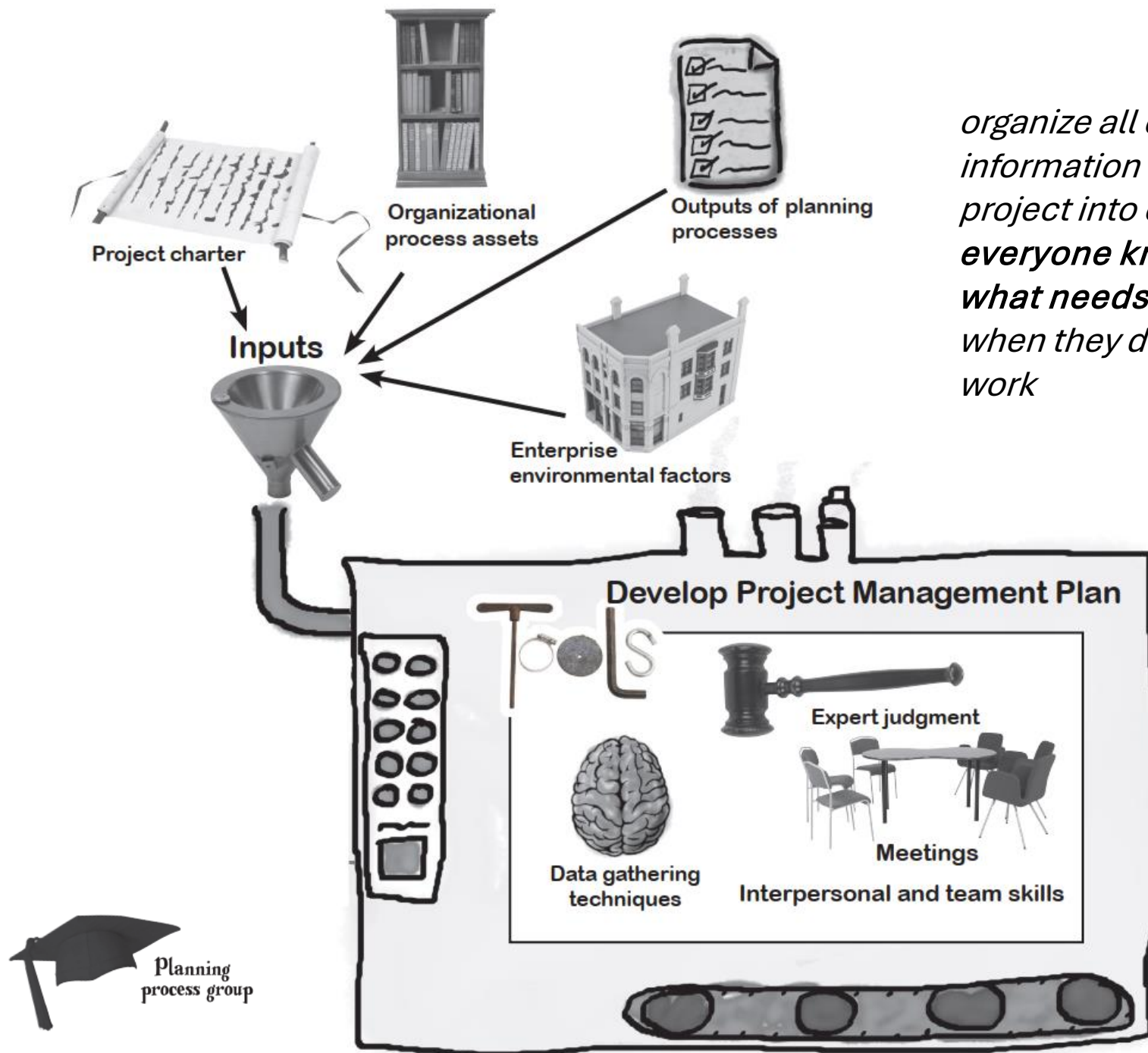
Tools & Techniques

- .1 Expert judgment
- .2 Data gathering
 - Brainstorming
 - Checklists
 - Focus groups
 - Interviews
- .3 Interpersonal and team skills
 - Conflict management
 - Facilitation
 - Meeting management
- .4 Meetings

Outputs

- .1 Project management plan





organize all of the information about your project into one place, so everyone knows exactly what needs to happen when they do the project work

Inputs

Project charter

- The high-level boundaries of the project

Outputs from processes

- Output from other **planning processes**
- Outputs from many of the **other processes**
- In addition, updates to these documents can necessitate updates to the project management plan

Enterprise Environmental Factors

- Governmental or industry standards
- Project management information system...

Organizational Process Assets

- Standardized Guidelines
- Project management plan template
- Change control procedures
- Configuration management knowledge base....

Output

Project Management Plan

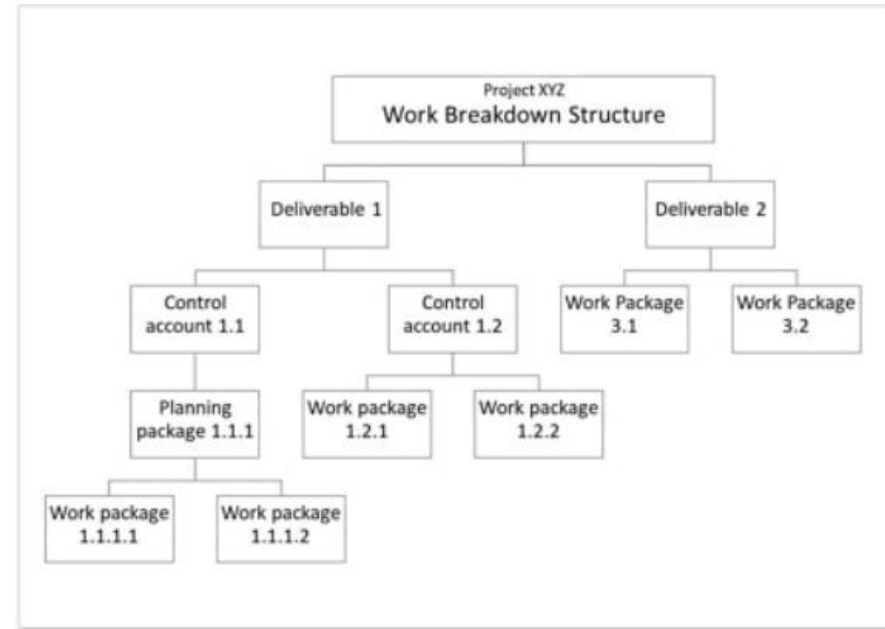
- The project management plan defines **how** the project is executed, monitored and controlled, and closed.
 - It integrates and consolidates all of the **subsidiary plans** and **baselines** from the planning processes.
 - The project management plan is **progressively elaborated** by updates, and controlled and approved through the Perform Integrated Change Control process.
- **Subsidiary plans**
 - Stakeholder management plan
 - Communications management plan
 - Scope management plan
 - Requirements management plan
 - Schedule management plan
 - Cost management plan
 - Quality management plan
 - ...
 - **Project baselines**
 - Scope baseline
 - Schedule baseline
 - Cost baseline
 - **Additional components**
 - Change management plan
 - Configuration management plan
 - ...

Subsidiary plans

- The **Scope Management plan** describes how scope changes are handled—like what to do when someone needs to add or remove a feature for a service or product your project produces.
- The **Requirements Management plan** describes how you'll gather, document, and manage the stakeholders' needs, and how you'll meet those needs with the project deliverables.
- The **Schedule Management plan** shows you how to deal with changes to the schedule, like updated deadlines or milestones.
- The **Cost Management plan** tells you how you'll create the budget, and what to do when your project runs into money problems.
- The **Quality Management plan** deals with problems that could arise when a product doesn't live up to the customer or client's standards.
- The **Resource Management plan** deals with changes in your staff, and to identify and handle any additional staffing needs and constraints you might have in your specific project.
- The **Communications Management plan** lists all of the ways that you communicate with your project's team, stakeholders, sponsors, and important contacts related to the project.
- The **Risk Management plan** is about detailing all the bad things that might happen and coming up with a plan to address each risk when and if it occurs.
- The **Procurement Management plan** focuses on dealing with vendors outside of your company.
- The **Stakeholder Management plan** focuses on managing the expectations of all of the people who are affected by the project.

Baseline

- The **scope baseline** is a snapshot of the scope, which helps you **keep track of changes** to the work that you'll be doing and the **planned deliverables** you'll be building.
- The **schedule baseline** does the same for the project schedule
- The **cost baseline** does the same for the budget



Below is a whole crop of factors the project team members discovered as they executed the project. Write down which subsidiary plan you'd look in to get some help.

1. The teachers want to go Bali, but Acme Travel doesn't book flights there so you need to subcontract one leg of the travel to another travel agency.

2. After reading online reviews, the teachers want to stay at better hotels. They tell you to increase their budget by 15% to do that.

3. Just as you're about to mail off the teachers' tickets, you notice they've been printed incorrectly.

4. The teachers might run into more bad weather, and you've got to figure out what contingencies you can put into place if that happens.

5. The teachers are concerned that they won't be able to get in touch with you when they're away.

6. One of the teachers realizes that he needs to come back earlier, and you want to make sure the budget reflects his lessened costs.

7. You find out that you need to get the tickets out earlier than expected, because the teachers' contract requires that all trips be preapproved by the superintendent of their school district.

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Procurement Management plan

2. After reading online reviews, the teachers want to stay at better hotels. They tell you to increase their budget by 15% to do that.

Cost Management plan

3. Just as you're about to mail off the teachers' tickets, you notice they've been printed incorrectly.

Quality Management plan

4. The teachers might run into more bad weather, and you've got to figure out what contingencies you can put into place if that happens.

Risk Management plan

5. The teachers are concerned that they won't be able to get in touch with you when they're away.

Communications Management plan

6. One of the teachers realizes that he needs to come back earlier, and you want to make sure the budget reflects his lessened costs.

Cost Management plan

7. You find out that you need to get the tickets out earlier than expected, because the teachers' contract requires that all trips be preapproved by the superintendent of their school district.

Schedule Management plan

Project Management Plan	Project Documents	
1. Scope management plan	1. Activity attributes	19. Quality control measurements
2. Requirements management plan	2. Activity list	20. Quality metrics
3. Schedule management plan	3. Assumption log	21. Quality report
4. Cost management plan	4. Basis of estimates	22. Requirements documentation
5. Quality management plan	5. Change log	23. Requirements traceability matrix
6. Resource management plan	6. Cost estimates	24. Resource breakdown structure
7. Communications management plan	7. Cost forecasts	25. Resource calendars
8. Risk management plan	8. Duration estimates	26. Resource requirements
9. Procurement management plan	9. Issue log	27. Risk register
10. Stakeholder engagement plan	10. Lessons learned register	28. Risk report
11. Change management plan	11. Milestone list	29. Schedule data
12. Configuration management plan	12. Physical resource assignments	30. Schedule forecasts
13. Scope baseline	13. Project calendars	31. Stakeholder register
14. Schedule baseline	14. Project communications	32. Team charter
15. Cost baseline	15. Project schedule	33. Test and evaluation documents
16. Performance measurement baseline	16. Project schedule network diagram	
17. Project life cycle description	17. Project scope statement	
18. Development approach	18. Project team assignments	

Tools & Techniques - Expert Judgment

- **Tailoring the project management process** to meet the project needs, including the dependencies and interactions among those processes and the essential inputs and outputs;
- **Developing technical and management details** to be included in the project management plan;
- **Developing additional components** of the project management plan if needed;
- Determining which project documents will be subject to **the formal change control process**; and
- Defining the level of **configuration management** to apply on the project



Tools & Techniques – Data gathering

Brainstorming

- Brainstorming is frequently used when developing the project management plan to gather ideas and solutions about the project approach.

Checklists

- A checklist may guide the project manager to develop the plan or may help to verify that all the required information is included in the project management plan.



Focus groups

- Focus groups bring together stakeholders to discuss the project management approach and the integration of the different components of the project management plan.



Interviews

- Interviews are used to obtain specific information from stakeholders to develop the project management plan or any component plan or project document.

Tools & Techniques – Meeting

Project kick-off meeting

- Its purpose is to communicate the objectives of the project, gain the commitment of the team for the project, and explain the roles and responsibilities of each stakeholder.
- The project kick-off meeting is usually associated **with the end of planning** and the **start of executing**.



Tools & Techniques – Skills

Meeting Management

- ensure that the numerous meetings that are necessary **to develop, unify, and agree on the project management plan** are well run.

Facilitation

- ensures that there is effective participation, that participants achieve a mutual understanding, that all contributions are considered, and that conclusions or results have full buy-in according to the decision process established for the project

Conflict management

- may be necessary to bring diverse stakeholders into alignment on all aspects of the project management plan.

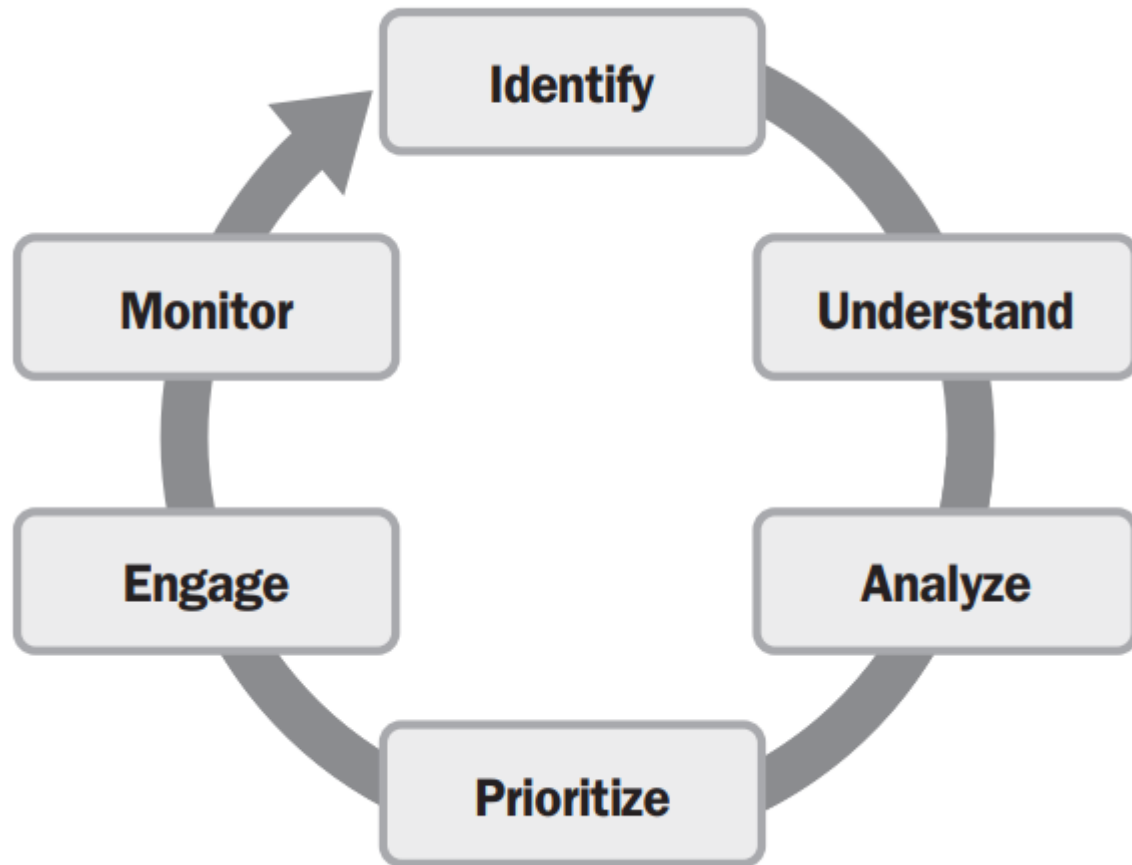


BUT IT WASN'T UNTIL
I MAPPED OUT ALL OF
THE PARTY GUESTS IN A
POWER/INTEREST GRID
THAT I REALLY KNEW
HOW TO KEEP THEM
SATISFIED.



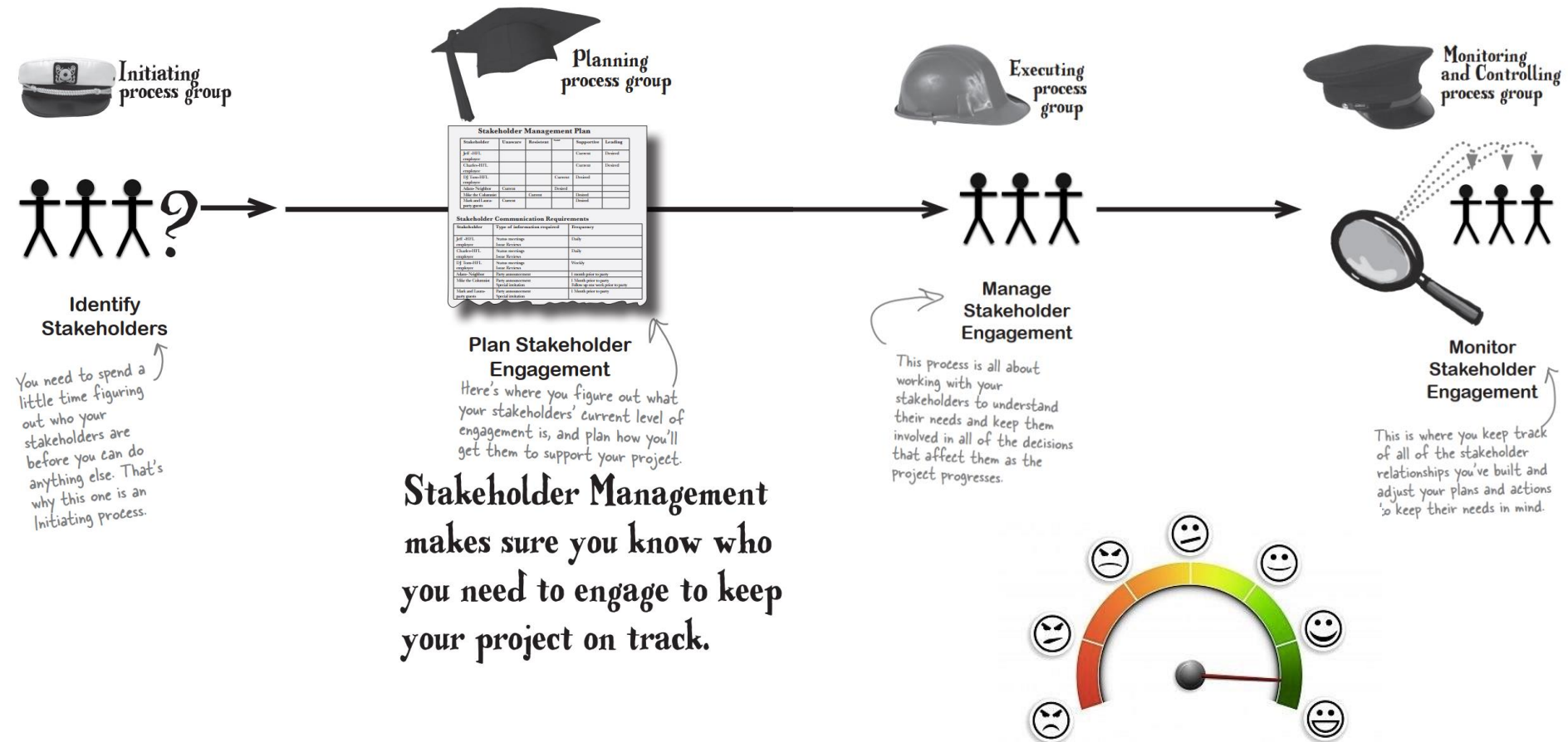
Stakeholder Management

Project management is about knowing your audience.



Active and proper stakeholder management => Lower the risks in your project

Stakeholder requirements and expectations sometimes change over the course of the project. Monitor Stakeholder Engagement makes sure you stay on top of those changes and adjust your plans accordingly.



Plan Stakeholder Engagement

Plan Stakeholder Engagement

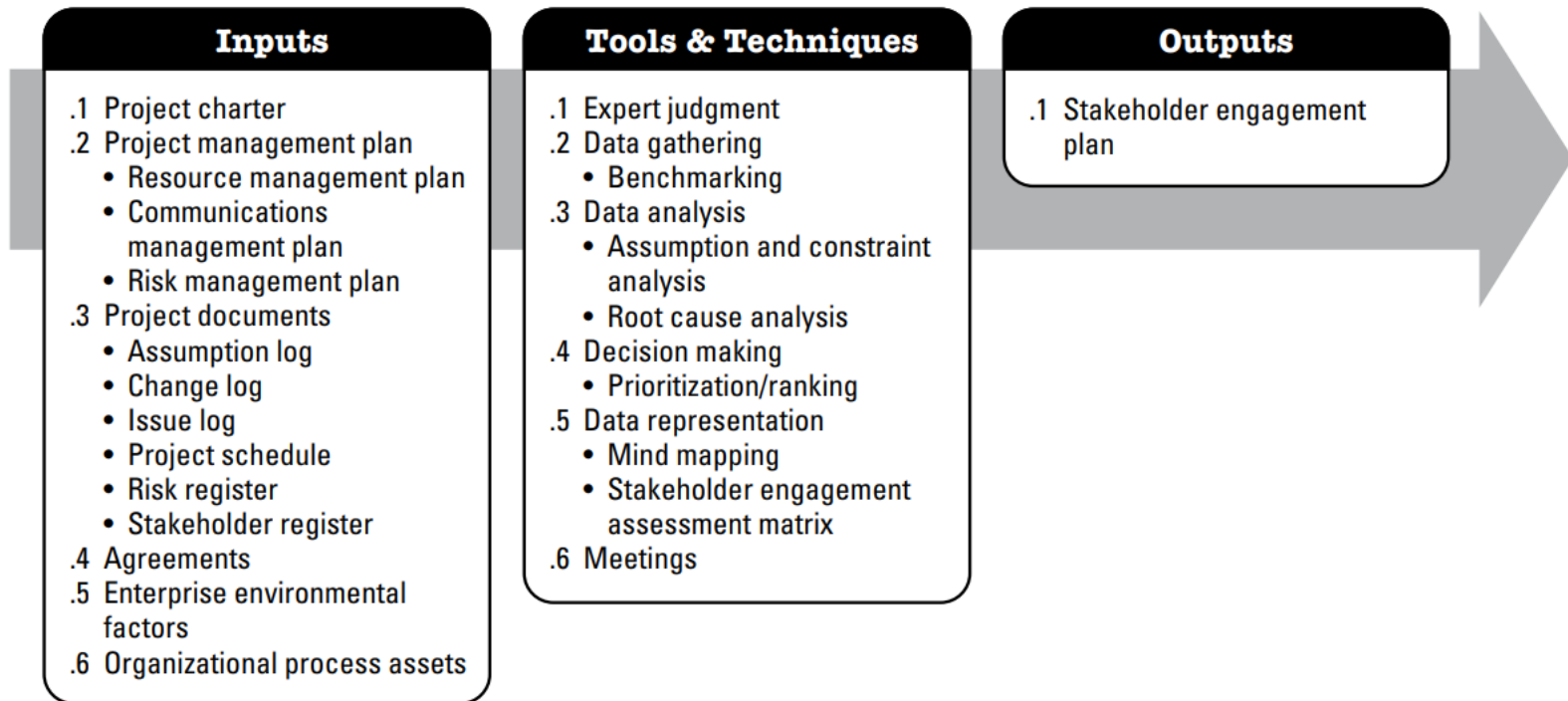
- **What:** developing **approaches to involve** project stakeholders based on their needs, expectations, interests, and potential impact on the project
- **Why:** provides an actionable plan to interact effectively with stakeholders
 - Stakeholder engagement is critical to project success
 - The levels of engagement of each stakeholder are different at each stage of the project
- **When:**
 - The start of a new phase of the project;
 - There are changes to the organization structure or within the industry;
 - New stakeholders, current stakeholders are no longer part of the stakeholder community, or the importance of particular stakeholders to the project's success changes
 - Outputs of other project process areas require a review of stakeholder engagement strategies.

Plan Stakeholder Engagement

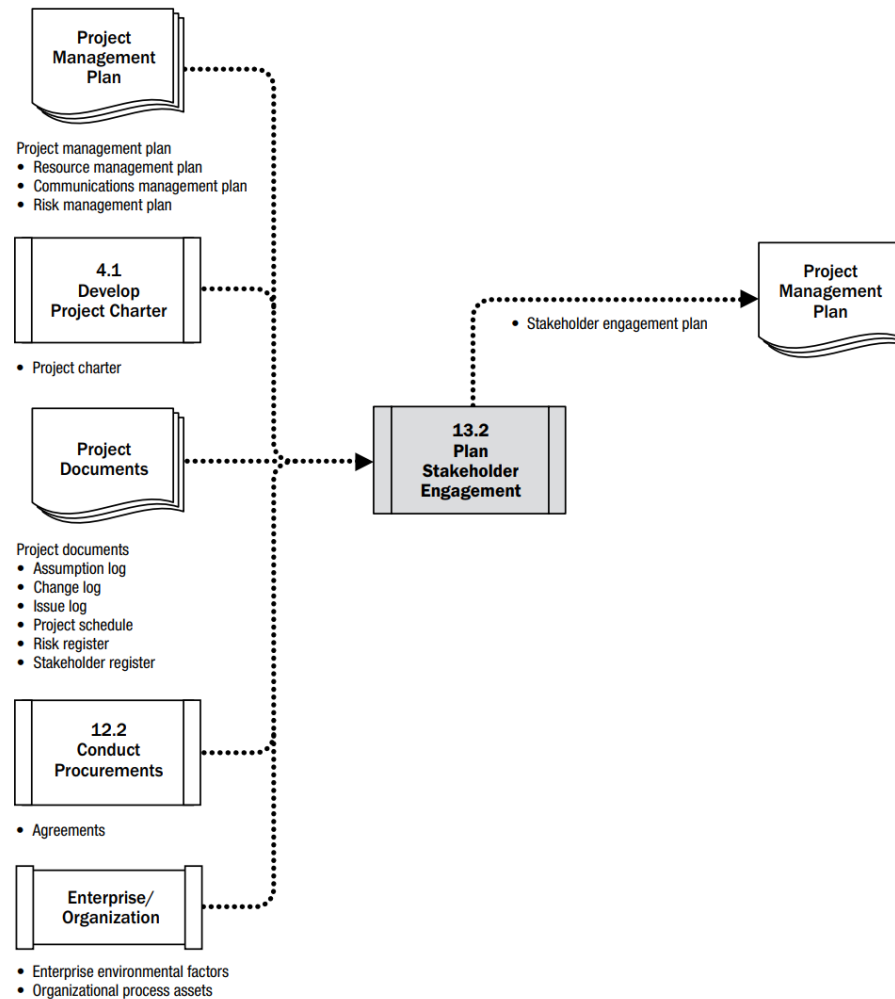
- **How?**

- Investigate the underlying reasons for the gap between current and desired level
- Develop the appropriate communication strategy to close the gap and effectively engage the stakeholder.

Process



Data Flow



Tools and Techniques

Stakeholder engagement assessment matrix

comparison between the current engagement levels of stakeholders and the desired engagement levels required for successful project delivery

Stakeholder	Unaware	Resistant	Neutral	Supportive	Leading
Stakeholder 1	C			D	
Stakeholder 2			C	D	
Stakeholder 3				D C	

'C' the current level of engagement.

'D' the desired level of engagement.

5 Levels of Stakeholder Engagement

Level 1

Unaware Stakeholders

- Unaware of the project
- Not clear on its implications
- Unaware of what it may impact

Level 2

Resistant Stakeholders

- Know about the project
- Know of its impacts
- Skeptical or opposed to it
- Don't support its goals or processes

Level 3

Neutral Stakeholders

- Aware of the project
- Don't feel strongly about it
- Neither for or against it

Level 4

Supportive Stakeholders

- Know about the project
- Have a positive attitude towards it
- Support its goals and work
- Willing to get behind it

Level 5

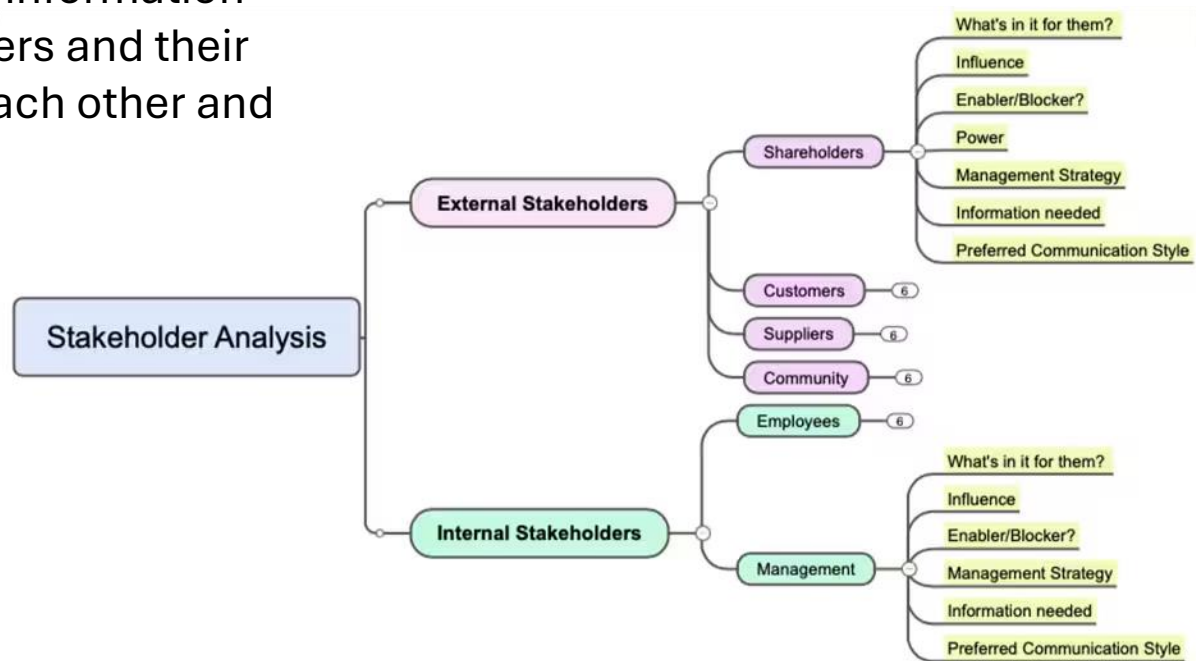
Leading Stakeholders

- Aware of the project
- Actively involved in it
- Champion the project to others
- Regularly take steps towards its success

Tools and Techniques

Mind mapping

Mind mapping is used to visually organize information about stakeholders and their relationship to each other and the organization.



Outputs

Stakeholder Engagement Plan

- Current and desired engagement levels of stakeholders
- Scope and impact of change to stakeholders
- Stakeholder communication requirements for the current phase of the project:
Format, content, level of detail, and language of information to be distributed to stakeholders, timeframe and frequency of distribution of information to stakeholders

[illegible]

Inputs

Project charter

- The project purpose, objectives, and success criteria

Project management plan

- **Resource management plan:** roles and responsibilities
- **Communications management plan:** need and techniques for communication among stakeholders.
- **Risk management plan:** risk thresholds or risk attitudes

Project Documents

- **Stakeholder register:** Stakeholders information and appropriate ways to engage
- Project schedule
- Risk register
- Issue log
- Change log
- Assumption log

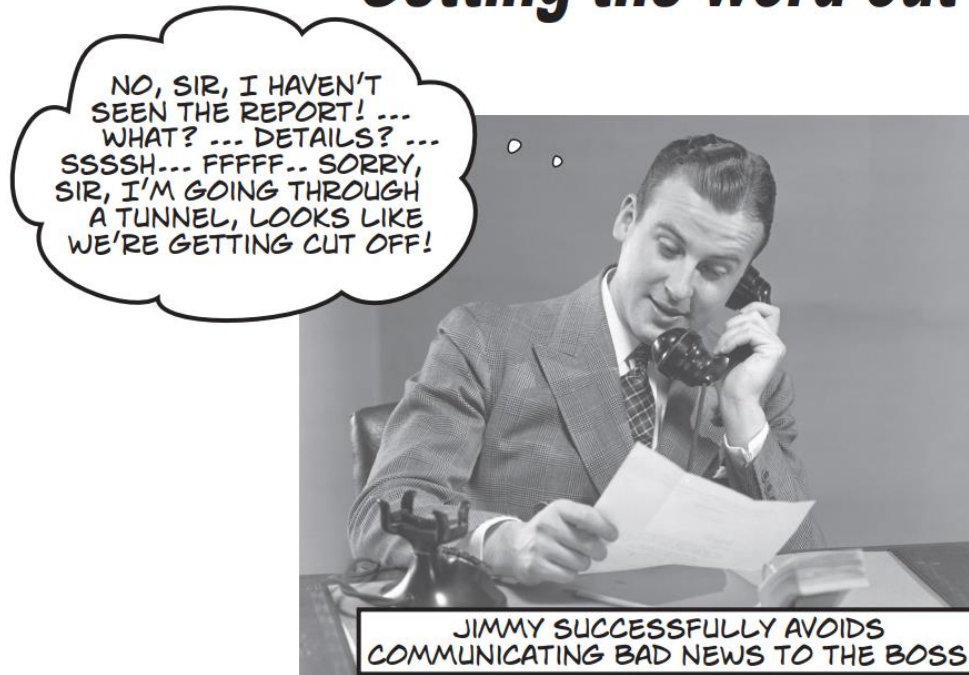
Agreements

- Contractor, suppliers

5. Enterprise environmental factors

6. Organizational process assets

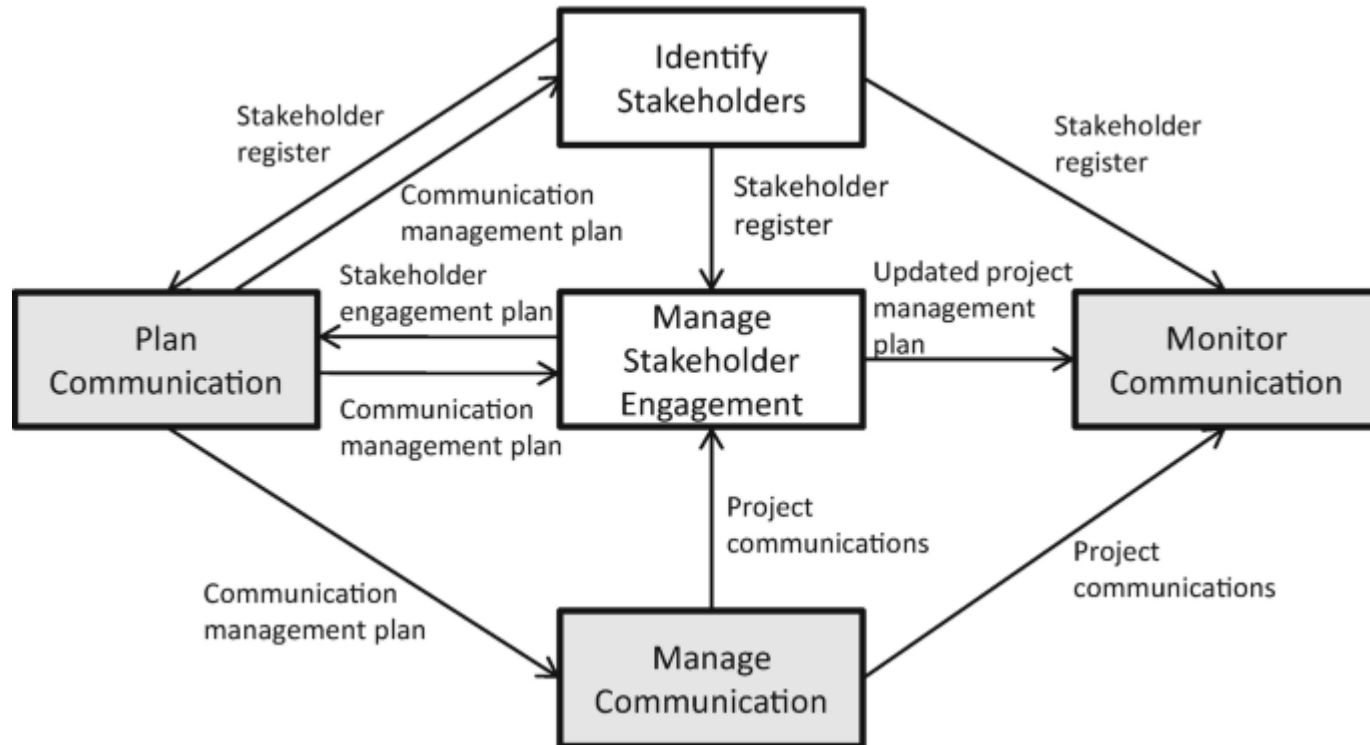
Getting the word out



Communication Management

Communications management is about keeping everybody in the loop

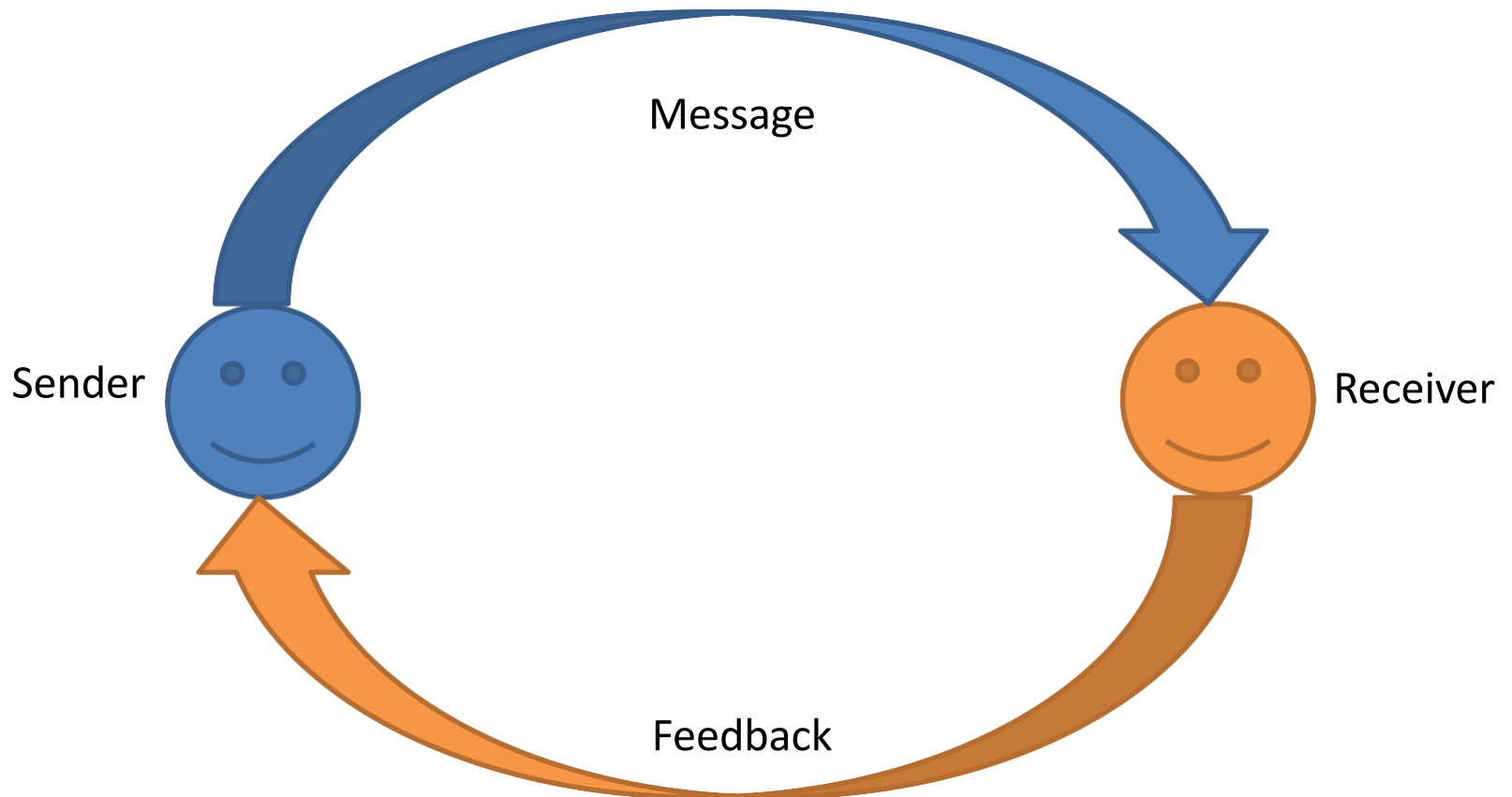
Stakeholder Mngt. & Communication Mngt.



*The project manager spends **90%** of his time in communication*

What is Project Communications?

Communication is the **exchange of information**, intended or involuntary. The information exchanged can be in the form of ideas, instructions, or emotions.



Dimensions of communication



Internal (within the project)



External (customer, vendors, other projects, organizations, the public);



Written (within the project)



Verbal (voice inflections)
and **nonverbal** (body language).

Dimensions of communication

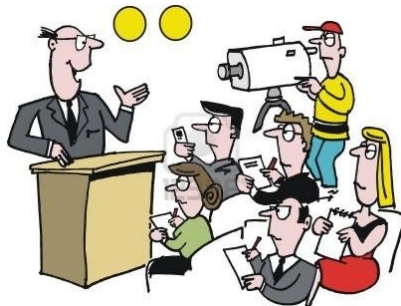


Formal (reports, minutes, briefings)



Informal (emails, memos, ad-hoc discussions);

Level of formality, structure, and format



Official (newsletters, annual report)



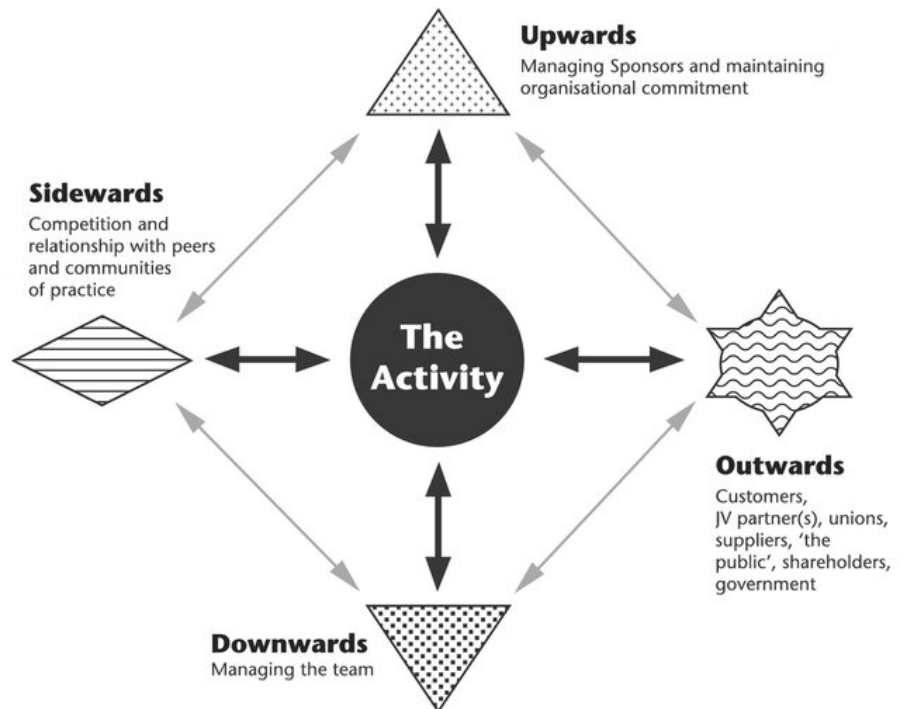
Unofficial (off the record communications);

Whether it represents the organization or not

Dimensions of communication

Hierarchical focus

- **Upward:** senior management, sponsor, steering committee
- **Downward:** team members or experts
- **Horizontal**
 - **Outward:** outside the project team(suppliers, government departments, the public, end-users, and regulators)
 - **Sideward:** the peers of the project manager such as other project managers or middle managers



Plan Communication Management

Plan Communication Management

- **What:** developing **an appropriate approach and plan** for project communications activities based on the **information needs** of each stakeholder or group, available organizational assets, and the needs of the project.
- **Why:** *effectively* and *efficiently* engage stakeholders by presenting relevant information in a *timely* manner
- **When:** performed throughout the project as needed

Effectively & Efficiently

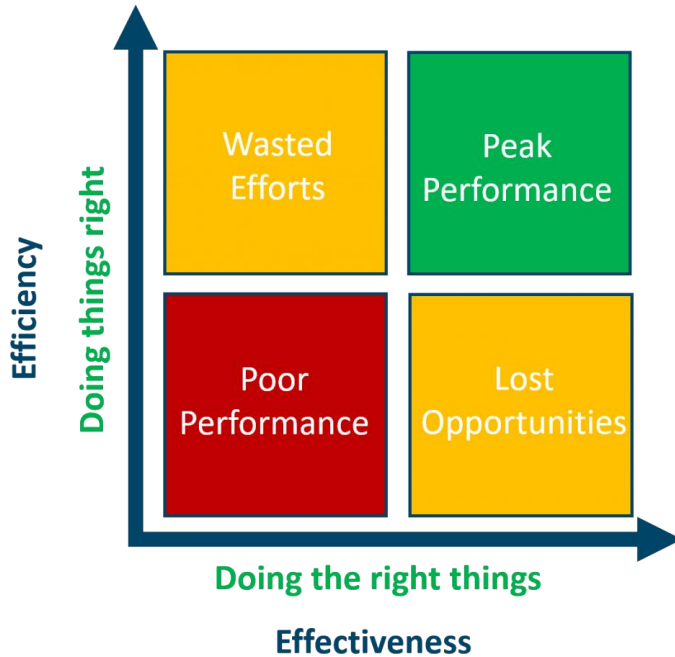
Productivity and optimization



Wrong audience



Delay in message delivery

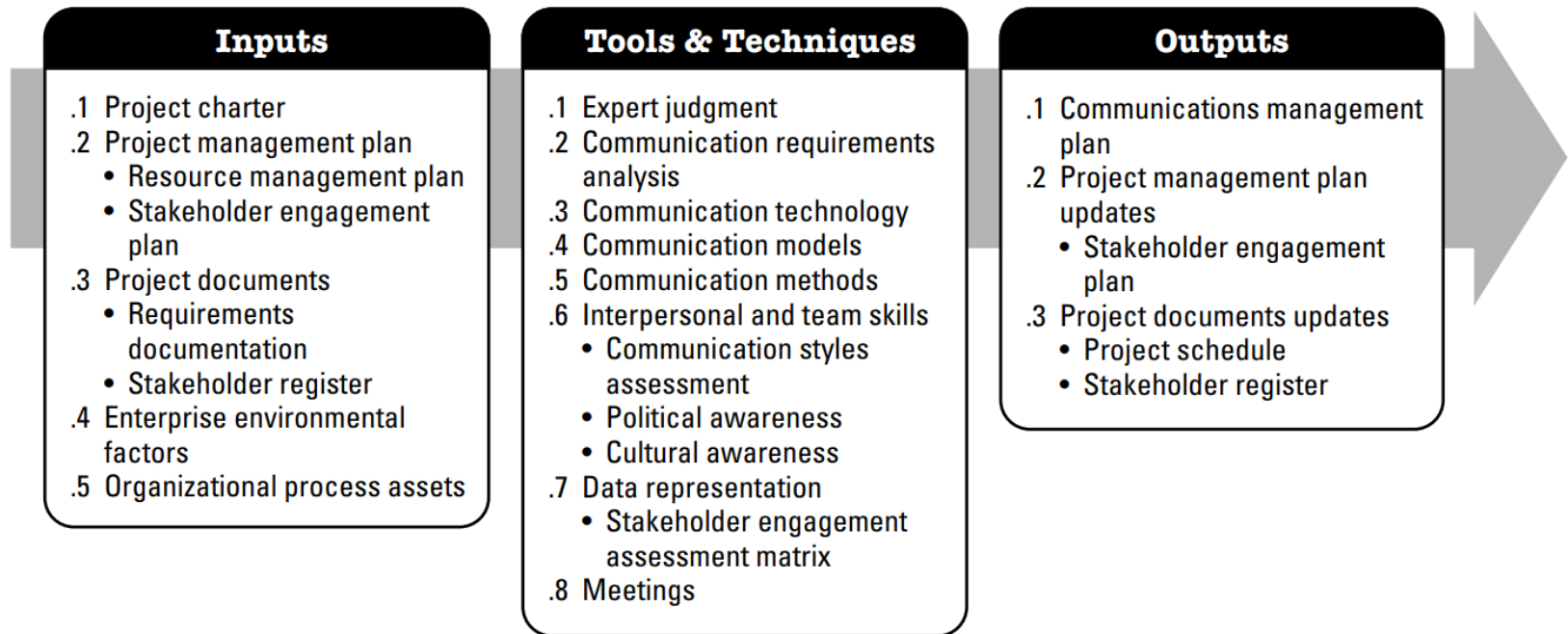


unrelevant/ Insufficient information

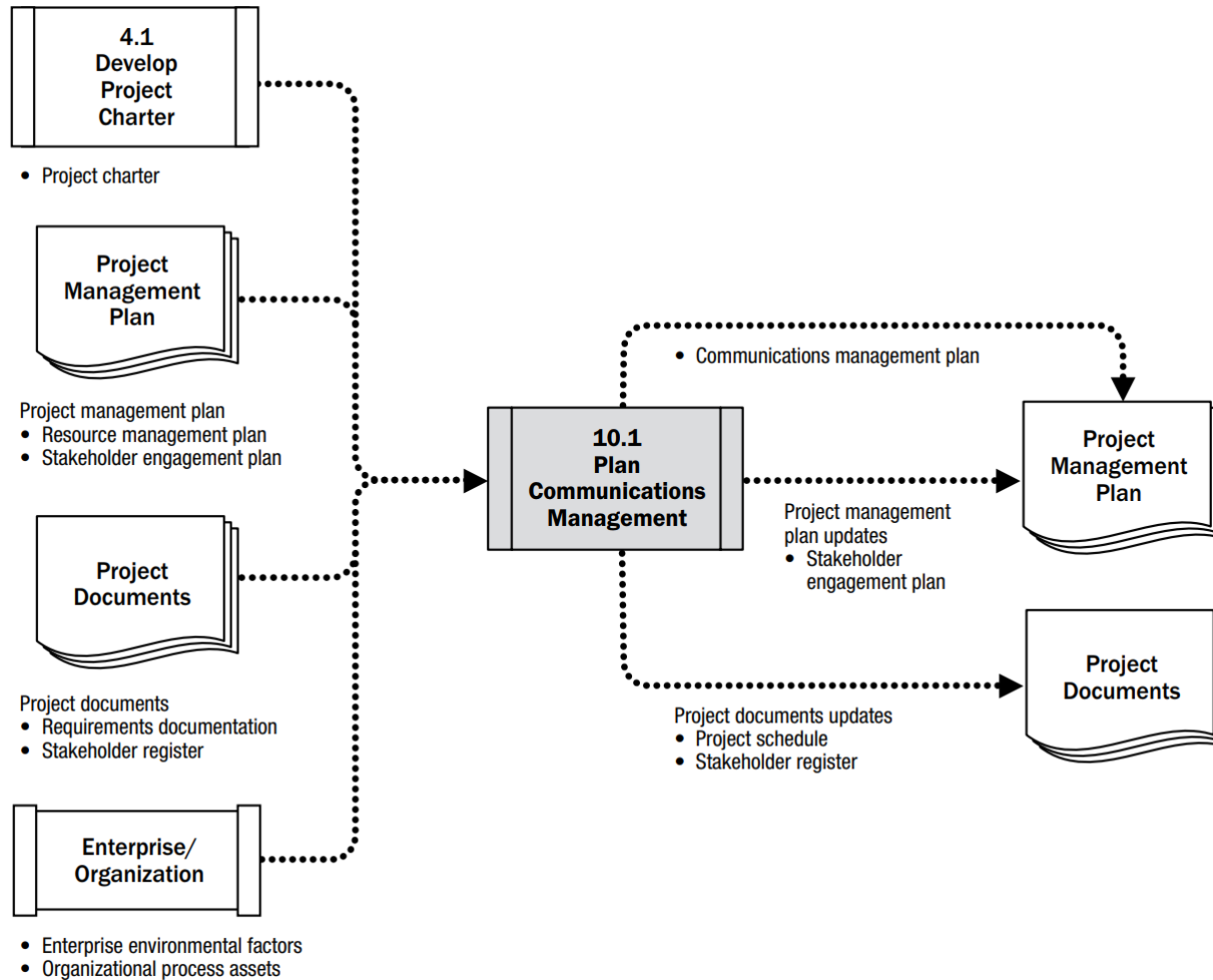


Results and impact

Process



Data Flow



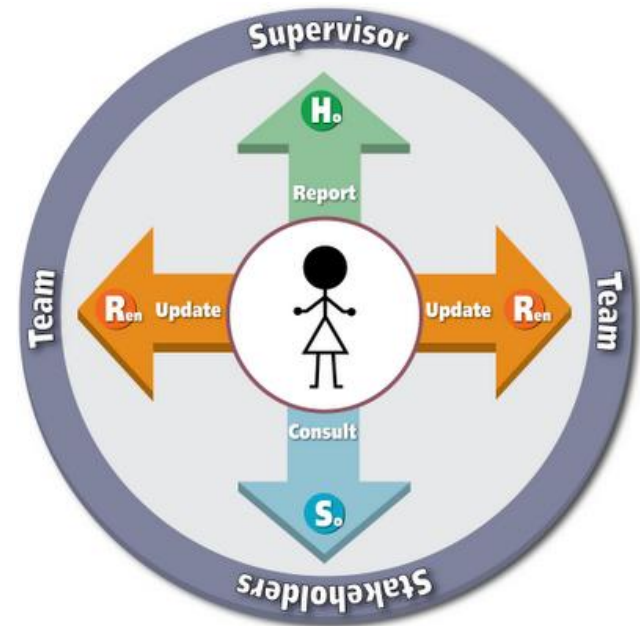
Tools & Techniques

Communication Requirement Analysis

- Stakeholder information and communication requirements (from the stakeholder register and stakeholder engagement plan)
- Project organization and stakeholder responsibility, relationships, and interdependence
- The format for communicating the information
- How often it's distributed,
- And who prepares it.

In addition, the methods of:

- Storage
- Retrieval
- And disposition of the project information



HOKOKU(報告)
REPORT TIMELY &
ADEQUATELY



RENRAKU(連絡)
COMMUNICATE
PROPERLY



SODAN(相談)
CONSULT

Tools & Techniques: Interpersonal and team skills

Communication styles assessment

Often used with unsupportive stakeholders, this assessment may follow a stakeholder engagement assessment to identify gaps in stakeholder engagement that require additional tailored communication activities and artifacts.

Political awareness

Recognition of power relationships, both formal and informal, and also the willingness to operate within these structures.

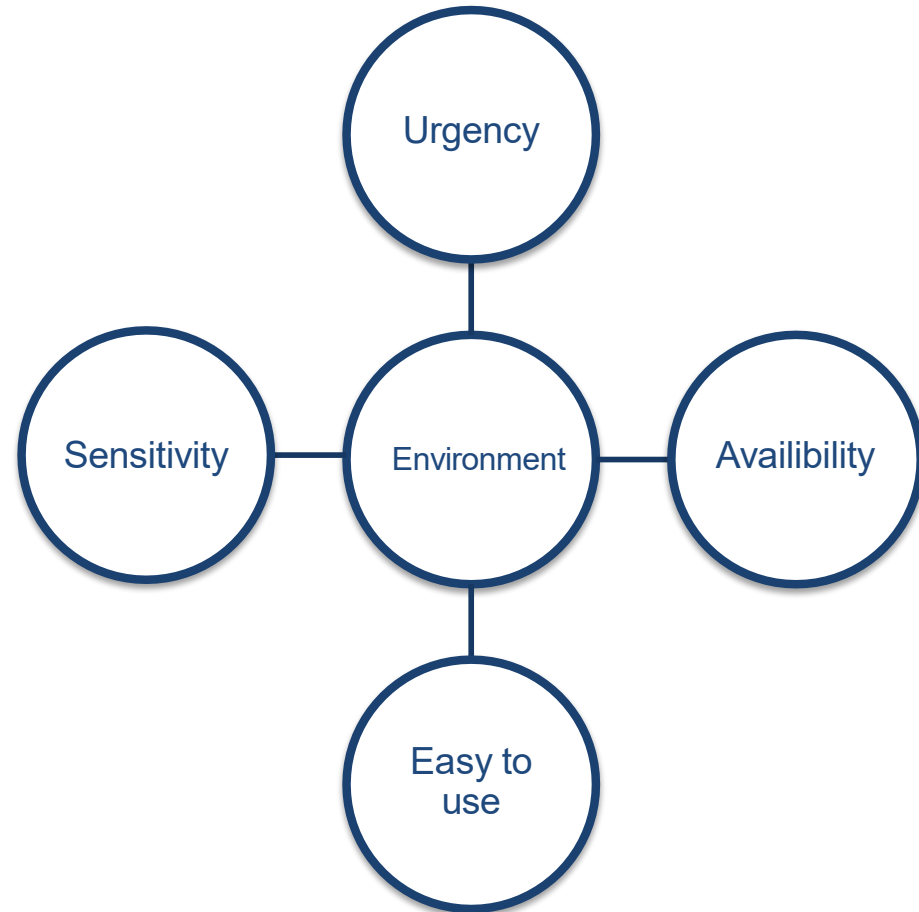
Cultural awareness

Minimize misunderstandings and miscommunication that may result from cultural differences.

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Stakeholder 1	C			D	
Stakeholder 2			C	D	
Stakeholder 3				D C	

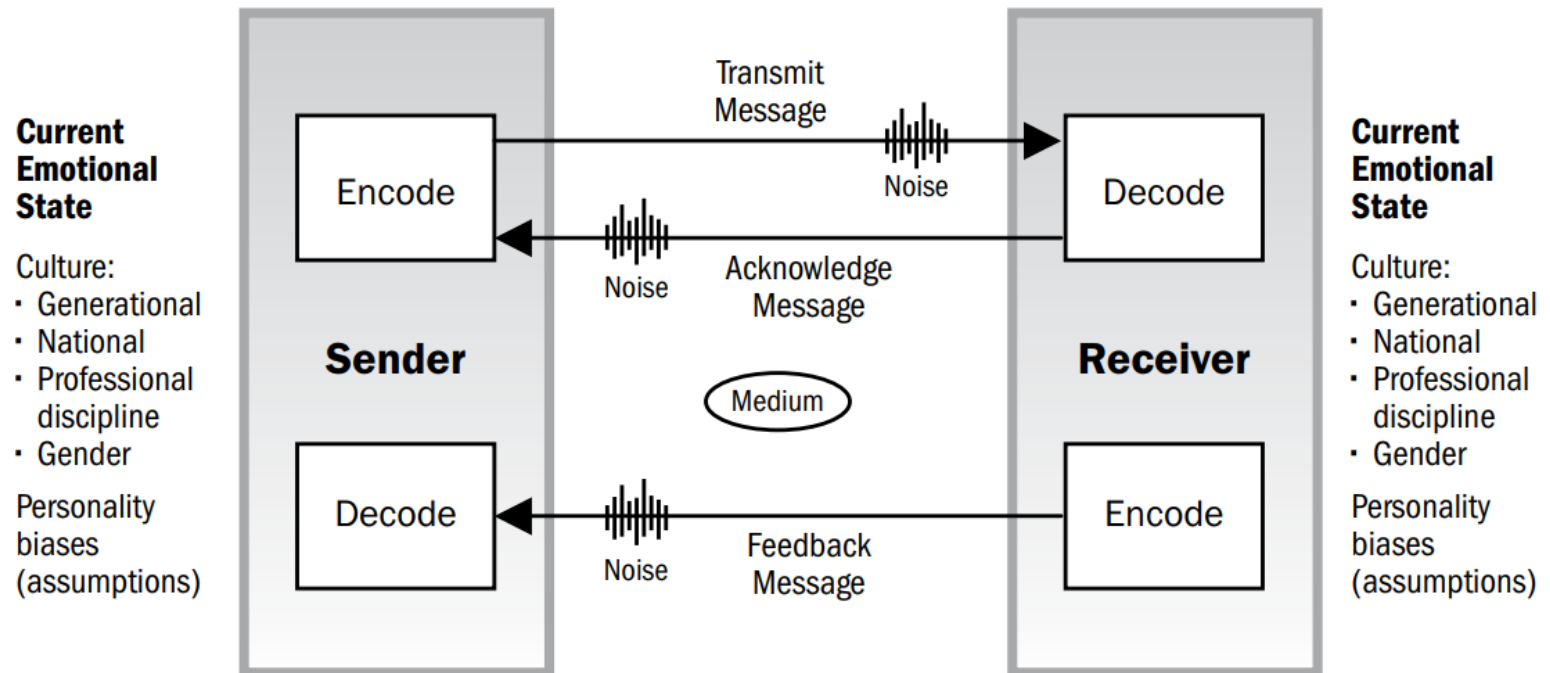
Communication Technology

- As part of the communications planning, the project manager should identify all of the required and approved methods of communicating.
- Communication modalities can also include meetings, reports, memos, e- mails, and so on.



Communication model

- Communication model is appropriate for the project that is undertaken and that any barriers (noise) are identified and managed.
- Communication blockers: noise surrounding, distance, improper encoding of messages, negative attitude, language, culture.....



Communication model

The sender's responsibility

- Encode the message clearly
- Select a communication method
- Send the message
- Confirm that the message was understood by the receiver



The receiver's responsibility

- Make sure that the information is received in its entirety,
- Decode the message
- Confirm that the message was understood



Communication methods

- **Push communication**

- Sent a specific information to specific recipients. Examples are letters, memos, e-mails, faxes etc.
- This ensures that message is sent but will not certify that it is actually received or understood.



- **Pull communication**

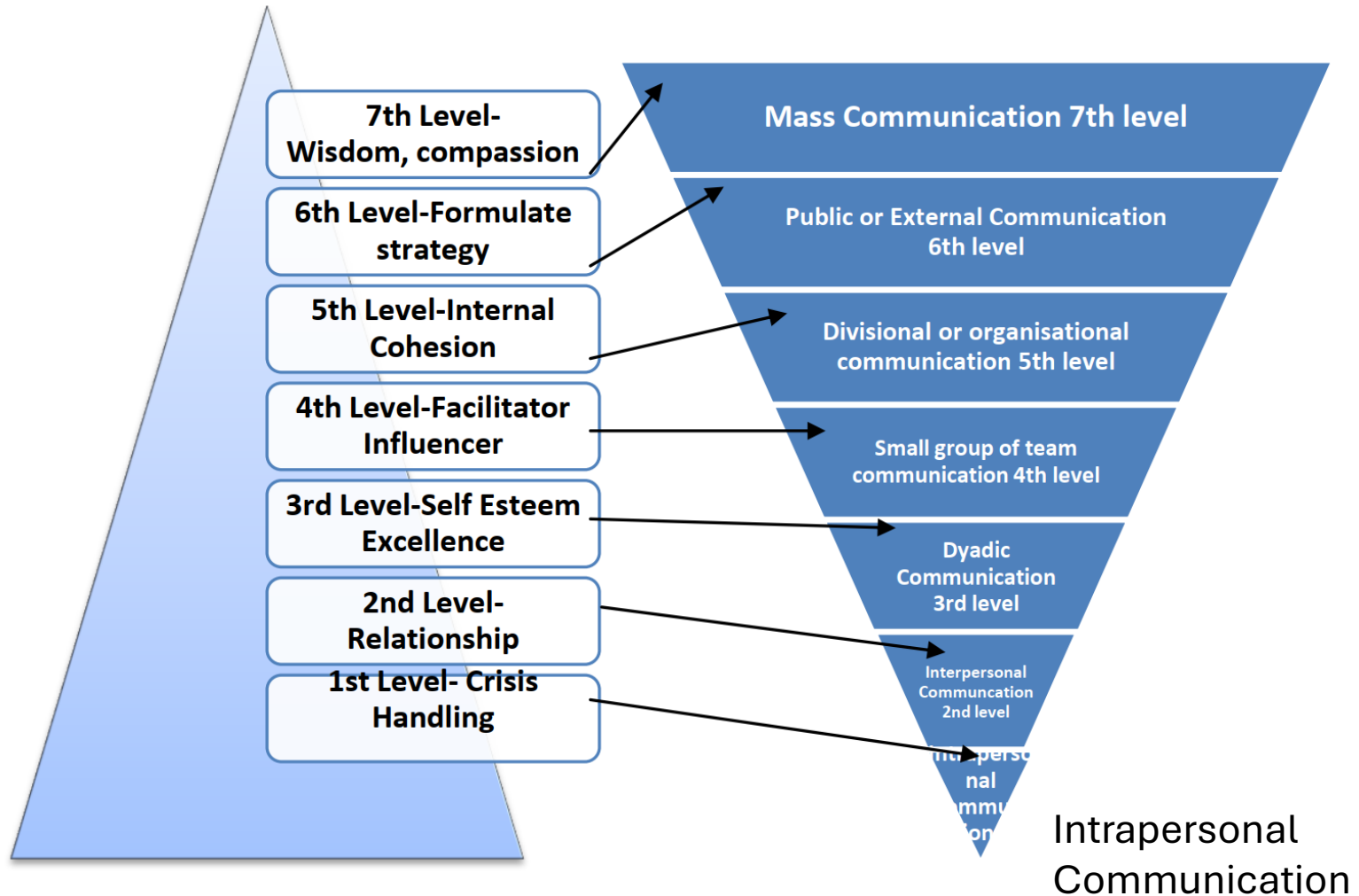
- Used for large volume of information and large audiences. Examples are internet sites, company data bases, e-learning etc.
- Recipients has to access the communication content.

- **Interactive communication**

- Information are pushed and pulled in real time between two or more parties in multidirectional (Phone calls, meetings, interviews...).



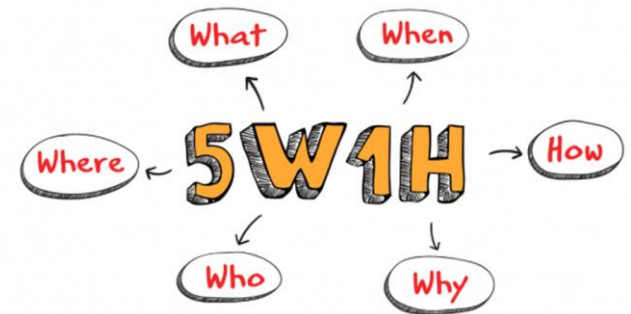
Communication levels



Outputs

Communications Management Plan

- Can be formal or informal,
- Highly detailed or broadly framed,
- Based on the needs of the project.
- Can include guidelines and templates for p status meetings, project team meetings, e- and email messages...



Project Management Plan Updates

- Stakeholder engagement plan

Project Document Updates

- Project schedule
- Stakeholder register: Stakeholder management strategy.

PROJECT COMMUNICATION PLAN TEMPLATE EXAMPLE

COMMUNICATION PLAN: SOFTWARE DEVELOPMENT TEAM

SUMMARY

This is the communication plan for the Atlanta software development team. It covers our schedule of meetings, where we store our shared documents, and how we communicate at other times.

COMMUNICATION GOALS

- Keep each other informed about the status of tasks (i.e., green, yellow, or red).
- Ask for and offer help where needed.
- Track budget to actual.
- Help groom the product backlog.
- Define information so you can share it with product owners and other stakeholders.

STAKEHOLDER INFORMATION

PERSON	ROLE / TITLE	CONTACT INFORMATION	COMMUNICATION FREQUENCY	FORMAT / CHANNEL	NOTES
Maria Hernández	Team Lead		Daily, weekly, monthly	The team lead facilitates the daily, in-person Scrum meeting, does weekly progress reports using Jira, and sends monthly, high-level timeline/budget/progress updates to the product owner by email.	Go-to for problem solving and questions
Jordan Oaks	Team Member - Programmer		Weekly	The team member/programmer participates in daily and weekly check-in meetings and emails.	Specializes in mobile
Wilbur Reynolds	Team Member - Programmer		Daily	The team member/programmer participates in daily and weekly check-in meetings and emails.	Oversees product backlog
David Runningbear	QA		Daily	QA reports on testing status and bug squashing.	
Sal Fiore	Release Manager		Weekly and as needed	The release manager updates release documentation.	Only person authorized to change release plan
Jennifer Planck	Product Owner		Monthly	The product owner reports team progress to leadership stakeholders monthly.	Not involved in daily Scrum meeting

COMMUNICATION TYPES

TYPE	WHEN / WHERE / PARTICIPANTS
DAILY SCRUM MEETING	We gather in person around the Scrum board for a stand-up meeting of 15 minutes.
SHARE	
• Each person reports on doing, done, to do. • Problems are flagged. • Ask for help if needed.	

TYPE	WHEN / WHERE / PARTICIPANTS
SLACK CHANNEL	This is open-ended, real-time communication and file sharing.
SHARE	
• Include the burndown chart. • Include documentation. • Ask questions so everyone can see.	

TYPE	WHEN / WHERE / PARTICIPANTS
SPRINT PLANNING AND RETROSPECTIVE	At the start and end of each two-week sprint, we meet to discuss what will be accomplished or to analyze the successes and failures of the past sprint.
SHARE	
• Share two days in advance: ◦ Agenda for meeting	